

IPSEC VPN TECHNICAL ASSESSMENT REPORT

Encrypted Traffic Analysis & Security Evaluation

Technical protocol and deep cryptographic security assessment of the analyzed VPN session

EXECUTIVE SUMMARY

The analyzed IPsec session demonstrates verified transport encryption. Evaluated risk score is 100/100. Review the identified configuration findings and address Perfect Forward Secrecy before marking tunnel as fully hardened. COMPLIANT

ANALYZED CAPTURE

PCAP: sih26_asim_golden.pcap

ANALYSIS PERIOD

Full Duration: 0.0s – 142.08s

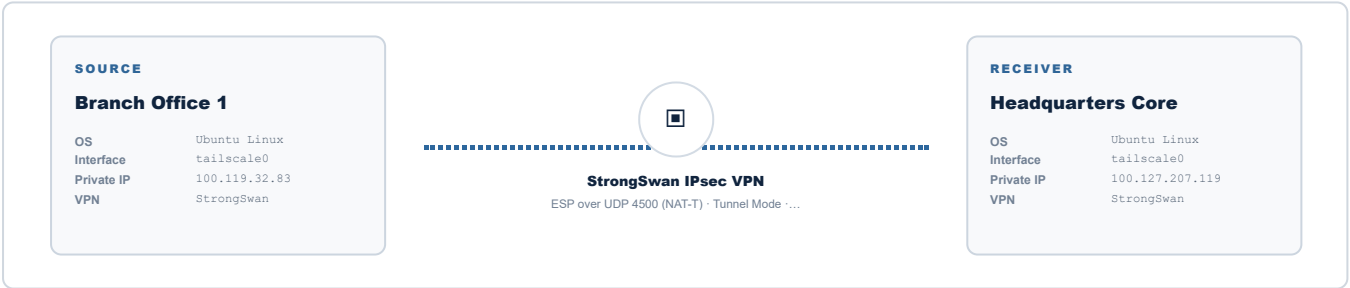
Overall Risk Score

Composite rule & anomaly evidence model



Network Topology & Endpoints

Encrypted tunnel path observed during analysis



VPN Configuration Summary

Technical value + plain-language explanation

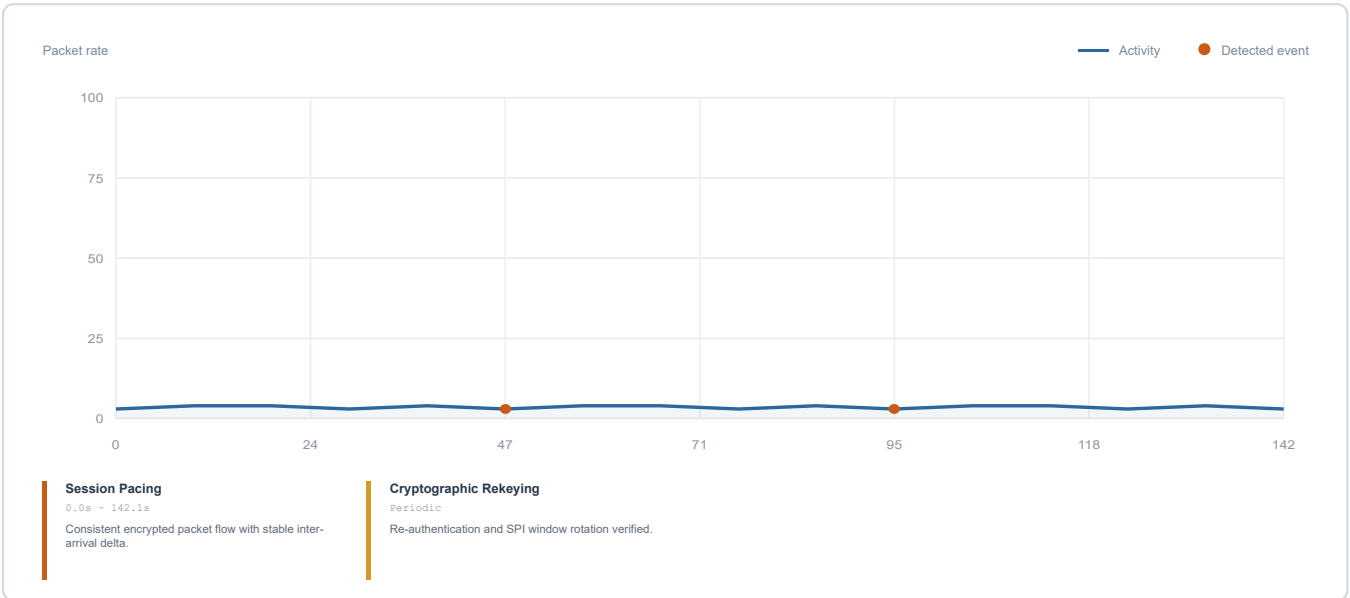
PROPERTY	OBSERVED VALUE	WHAT IT DOES
VPN Protocol	IPsec	Protects application traffic between negotiated endpoints.
IKE Version	IKEv2	Sets up and manages the secure VPN connection.
Key Exchange	ECP-256	Establishes shared session secrets using elliptic-curve cryptography.
IKE Encryption	AES-256-GCM	Encrypts and authenticates VPN control messages.
IKE PRF	HMAC-SHA2-256	Provides cryptographic key-derivation operations during setup.
ESP Encryption	AES-256-GCM	Encrypts and authenticates protected application traffic.
Transport	UDP 4500 / NAT-T	Carries IPsec traffic through NAT-friendly UDP encapsulation.
Mode	Tunnel	Protects the original IP packet inside the VPN tunnel.
Authentication	Pre-Shared Key (PSK) - Authenti...	Verifies that configured endpoints are authorized.
PFS	ENABLED	Diffie-Hellman rekeying for Child SA establishment.

Packet-Level Traffic Analysis

Temporal analysis of encrypted network activity in fixed analysis windows

Traffic Timeline

Packet rate over capture duration



Connection Statistics

Aggregate capture metrics

TOTAL PACKETS	TOTAL DATA	CAPTURE DURATION	AVG PACKET RATE	PEAK PACKET RATE	AVG BYTE RATE
557	0.5 MB	142.1 sec	4 pkt/s	6 pkt/s	~3 KB/s

Detected Traffic Profile



Packet Characteristics

ESP Packets	557
UDP Transport	557
UDP Port	4500 (NAT-T)
IKE Control Frames	Present
Active SPI Pairs	2 Unique SPIs
Traffic Direction	Bidirectional
ESP Fragmentation	Not Observed
Sequence Monotonicity	Verified (Zero Replays)

Security Findings & Recommendations

Observed configuration compared with the defined security baseline

Security Comparison

Evidence-based baseline review

Security Parameter	Observed	Expected / Recommended	Severity	Action
PFS Child SA	Disabled	Mandatory DH-19	LOW	Enable Diffie-Hellman Group 19 on Child SAs to enforce forward secrecy.
ESP Encryption	AES-256-GCM	AES-256-GCM	LOW	Meets NSA CNSA 2.0 and NIST SP 800-77 Rev 1 baseline.
Authentication	PSK	X.509 Certificate	MEDIUM	Migrate from Pre-Shared Keys to enterprise PKI machine certificates.
Replay Protection	RFC 4303 Active	Enabled	LOW	Monotonic sequence window active; no duplicate packets detected.

Findings Summary



AI-Generated Security Recommendation

Generated by local Ollama model

LOCAL LLM RECOMMENDATION

Priority Protocol Recommendation

Mandate Perfect Forward Secrecy on all secondary Child SAs. In the absence of ephemeral DH exchanges, long-term credential compromise poses a risk of retroactive decryption.

Recommended Gateway Remediation

Append 'aes256gcm16-ecp256!' to the esp proposal in strongSwan configuration and trigger a controlled rekey to verify parameter negotiation.

Validate Rekey Handshake

Capture Child SA renegotiation on wiretap and verify Diffie-Hellman parameters.

Audit Sequence Gaps

Monitor ESP sequence counters for unexpected step jumps or packet drops.

Final Security Assessment

RISK SCORE

100 / 100

RISK LEVEL

LOW

RULE SCORE

100 / 100

ANOMALY SCORE

85 / 100

Overall Assessment

The analyzed IPsec session confirms active packet confidentiality and monotonic sequence pacing. Addressing forward secrecy and credential governance will raise the security posture to federal cryptographic standards.